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جامعة الإمام عبد الرحمن بن فيصل
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ARTI 303 – Programming for AI
(Project Report)

Project Title: Building a Basic Sentiment Analyzer with Python and TextBlob

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Group: A2

Submission Date:

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1. Introduction

Customer reviews, especially on platforms like Amazon, are highly influential in customer decisions and product reputation. Sellers commonly rely on thousands of reviews to understand customer sentiment. Because manual analysis is time-consuming, sentiment analysis provides an automated way to classify reviews as positive, negative, or neutral.
This report presents the entire workflow of sentiment analysis using Python, Pandas, and TextBlob to analyze over **6,000 reviews of the Samsung Galaxy S4 smartphone.** Valuable insights are to be drawn from this, summarize customer satisfaction, and support decision-making for product improvement.

2. Problem statement

Amazon sellers face a massive number of customer reviews, making it difficult to manually analyze thousands of opinions. This process is time-consuming and prone to human error. Therefore, the challenge lies in developing an intelligent system that automatically categorizes reviews as positive, neutral, or negative, helping sellers quickly understand customer feedback.

3. AI Solution

The solution was implemented using Python, Pandas, TextBlob, and Matplotlib.
To analyze customer feedback, each review was processed using a custom function sentimentAnalyzer(text) that classifies the sentiment based on TextBlob polarity scores.

The system classifies the sentiment as:

- Negative: polarity < -0.2
- Neutral: $-0.2 \leq$ polarity ≤ 0.2
- Positive: polarity > 0.2

These rules were applied to every review in the dataset to generate a new Sentiment column.
In addition, a compute_polarity(text) function was used to extract the exact polarity score for each review.
To structure the solution more professionally, a simple SentimentAnalyzer class was also implemented. This class receives the review text, computes its polarity using TextBlob, and returns both the polarity value and the corresponding sentiment label.

4. Use case (the product you selected with the analysis results)

4.1 Dataset Filtering & Statistics

The dataset used is the 'Amazon Unlocked Mobile Phones' dataset containing more than 400,000 reviews. A filter was applied for all product names starting with “Samsung Galaxy S4” to avoid accessory items.

Dataset Statistics:

- Number of rows (reviews): 6056
- Number of columns: 7
- Shortest review length: 2
- Longest review length: 8188
- Average review length: 180.55003302509908

These values indicate a large, diverse dataset suitable for meaningful sentiment analysis.

Missing Values

A few missing values were found during dataset inspection, primarily in the Brand, Price, and Votes columns. These fields were not imputed or cleaned because they were not necessary for the sentiment analysis tasks.

The Review column also had a few missing entries. However, by transforming None or empty inputs into an empty string, the implemented SentimentAnalyzer class automatically handles missing review text. This guarantees that polarity scores are produced accurately, even in cases where some reviews are missing or insufficient.

4.2 Sentiment Results Overview

After applying the sentimentAnalyzer function, the distribution of sentiment was:

- Negative: 296
- Neutral: 2381
- Positive: 3379

This indicates that most customers expressed positive experiences with the Samsung Galaxy S4.

4.3 Visualizations

Sentiment Distribution

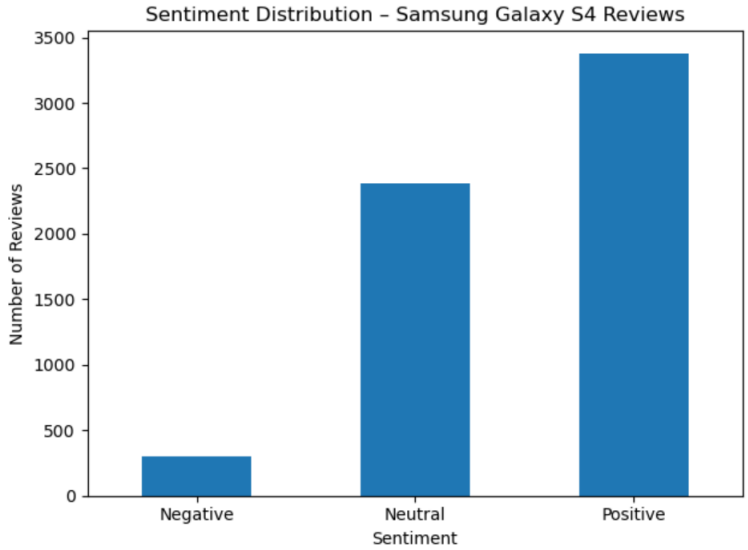

Figure 1: Distribution of Positive, Neutral, and Negative reviews.

Polarity Histogram

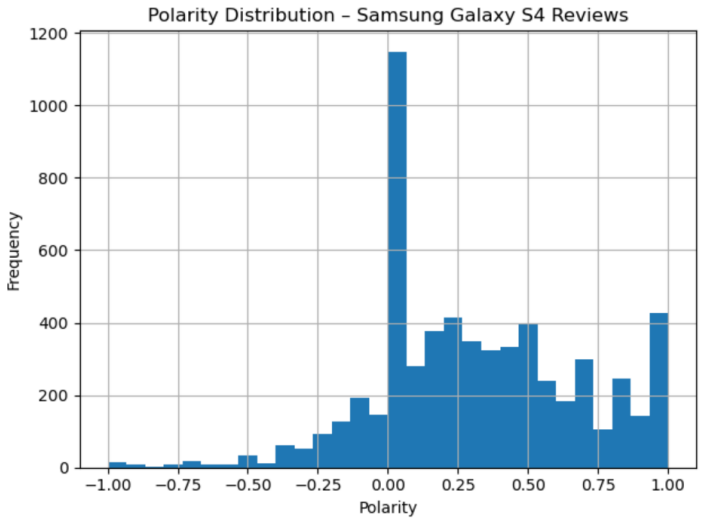

Figure 2: Histogram showing polarity distribution from -1 to +1.

4.4 Mismatched Reviews Analysis

A total of 214 mismatched cases were identified where the sentiment classification did not match the numerical rating.

Examples include:

- Rating 1 or 2, but sentiment classified as Positive
- Rating 4 or 5, but sentiment classified as Negative

This occurs due to mixed wording, sarcasm, complaints about seller instead of product, or inconsistent grammar affecting polarity evaluation.

5. Alignment between AI solution and client's needs*

The intelligent sentiment analysis system supports Amazon sellers by:

- Automatically classifying reviews into sentiment categories
- Summarizing customer satisfaction effectively
- Highlighting recurring issues found in negative reviews
- Providing data-driven insights to improve product quality
- Allowing fast evaluation without manually reading thousands of reviews

This aligns directly with the assignment requirements and the seller's needs.

6. Conclusion

The sentiment analysis system successfully processed over 6000 Samsung Galaxy S4 reviews and generated insights into customer satisfaction trends. The results indicate high overall satisfaction, with some recurring concerns identified. The system fulfills all ARTI 303 project requirements and demonstrates practical application of sentiment analysis in real business contexts.

Marks Distribution

CLO	Topic	Assessment	Dedicated Marks	Earned Mark
CLO3	Conditional Execution	Poor/Developing/Developed/Exemplary	1	
	Function, Modules, and Packages	Poor/Developing/Developed/Exemplary	3	
CLO4	Classes and Objects	Poor/Developing/Developed/Exemplary	4	
CLO5	Python DS: Pandas	Poor/Developing/Developed/Exemplary	6	
	Python DS: Visualization			
CLO6	PI-5.1: Students demonstrate the abilities to participate in team activities.	Poor/Developing/Developed/Exemplary	1	
		Total =	**15**	